Digital Model

Faltinsen linear potential theory · API 650 Annex E

Tank sloshing resonance explorer

A representative LNG-scale tank swept across fill level: the fundamental sloshing periods (prismatic and cylindrical, linear potential theory) and the API 650 Annex E convective period, read live against a vessel roll-period band. Every point is a live hydrodynamics.sloshing evaluation.

49 engine evaluations	**<0.5%** API 650 ↔ potential-flow agreement	**6.7–15.5 s** fundamental period vs fill

WHAT YOU GET

- Rectangular & upright-cylindrical fundamental sloshing periods from linear potential-flow theory (Faltinsen & Timokha), plus the API 650 Annex E convective period
- Live resonance flag against a representative vessel roll band — partial fill lengthens the sloshing period into the vessel-motion range, the coupling external diffraction (AQWA/OrcaWave) does not capture
- The analytical / class-simplified screening tier; CFD (VOF free-surface) is reserved for the violent/impact and resonant cases it flags

Ask Deckhand: “Run the digitalmodel tank sloshing resonance explorer and send me the report.” — send to @the_deckhand_bot to run it live.